ELEC4631 Solutions Tutorial 11

1.

$$
\begin{aligned}
\dot{V}(x) &= 2x_1\dot{x}_1 + 2x_2\dot{x}_2, \\
&= 2x_1(-x_1^3 + u) + 2x_2x_1, \\
&= -2x_1^4 + 2x_1u + 2x_2x_1.
\end{aligned}
$$

Choose u to cancel the indefinite signed term $2x_2x_1$, $u = -x_2$. So, $\dot{V}(x) = -x_1^4$, it is nsd around $(0,0)$. $\dot{V}(x_1,x_2) = 0$ whenever $x_1 = 0$. Any trajectory such that $\dot{V}(x_1,x_2) = 0$ along it satisfies $x_1(t) = 0$ and $\dot{x}_1(t) = 0$ $\forall t \geq 0$. This gives the ODE (after substituting $u = x_2$):

$$
\begin{aligned}
0 &= -x_2, \\
\dot{x}_2 &= 0,
\end{aligned}
$$

So, $x_2(t) = 0$ for all $t \geq 0$. We conclude that the only trajectory that stays in $I = \{x \in \Omega_{r,c} \mid \dot{V}(x) = 0\}$ when it starts in I is the constant trajectory $x(t) = 0$ for all $t \geq 0$. Thus, the largest positively invariant set M inside I is just $M = \{(0,0)\}$. Since V is also radially unbounded, the global asymptotic stability of $(0,0)$ follows from the Krasovskii-LaSalle Invariance Principle.

2. For $\lambda > 0$, $V(x_1,x_2)$ is a positive definite function (hence, it is also lpd around $(0,0)$ but for global asymptotic stability the locality is not enough, we need the stronger property of positive definiteness). Computing $\dot{V}$ gives:

$$\dot{V}(x_1,x_2) = x_2\dot{x}_2 + \lambda x_1\dot{x}_1 = x_2(-x_2|x_2| + x_1^3 + u + \lambda x_1).$$

Setting the control law as $u = -x_1^3 - \lambda x_1 = -x_1(x_1^2 + \lambda)$ gives

$$\dot{V}(x_1,x_2) = -x_2^2|x_2|.$$

So, $\dot{V}$ is only nsd around the origin, rather than nd. To establish converge of trajectories to the origin requires Krasovskii-LaSalle. Note that since $\lambda > 0$, the control u vanishes only if $x_1 = 0$. If $\dot{V}(x_1, x_2) = 0$ along a trajectory then $x_2(t) = 0$ and $\dot{x}_2(t) = 0$ for all $t \geq 0$. In this case the ODE is (after substituting the expression for u):

$$\begin{aligned} \dot{x}_1 &= 0, \\ 0 &= \lambda x_1. \end{aligned}$$

Since $\lambda > 0$ this gives $x_1(t) = 0$ for all $t \geq 0$. That is, the only trajectory that stays in $I = \{x \in \Omega_{r,c} \mid \dot{V}(x) = 0\}$ when it starts in I is the constant trajectory $x(t) = 0$ for all $t \geq 0$. Thus, the largest positively invariant set M inside $I = \{(x_1, x_2) \in \Omega_{r,c} \mid \dot{V}(x_1, x_2) = 0\}$ is just $M = \{(0, 0)\}$. Since V is also radially unbounded, the global asymptotic stability of $(0, 0)$ follows from the Krasovskii-LaSalle Invariance Principle.

3. Directly from the lecture notes, we have the following candidate control law ($\gamma_1 = 1$, $\gamma_2 > 0$)

$$u = -mgl \sin x_1 - x_1 - \gamma_2 x_2,$$

giving

$$\dot{V}(x) = -(b + \gamma_2) x_2^2.$$

Since $\dot{V}(x)$ is only nsd around the origin, to establish $x(t) \to 0$ as $t \to \infty$ requires Krasovskii-LaSalle. Proceed as in the solutions for Q1 and Q2 above and use the radial unboundedness of V to prove that the origin is globally asymptotically stable.

4. (i) $V(x, y) = 0$ when $(x, y) \in C$ otherwise $V(x, y) > 0$ for $x \notin C$.

(ii)

$$\begin{aligned} \dot{V}(x, y) &= (x^2 + y^2 - 1)(x\dot{x} + y\dot{y}) \\ &= -(x^2 + 5y^2)(x^2 + y^2 - 1)^2. \end{aligned}$$

So, $\dot{V}(x, y) = 0$ if $(x, y) = (0, 0)$ or $(x, y) \in C$, otherwise $V(x, y) < 0$.

(iii) $\Omega_{r,c}$ contains C but does not contain the origin $(0, 0)$. We apply the K-L invariance principle to show that the trajectories starting on $\Omega_{r,c}$ converges to a point in C. We need to show that if a trajectory of the system starts in C it will stay in C, showing that the largest positively M in $I = \{(x, y) \in \Omega_{r,c} \mid \dot{V}(x, y) = 0\} = \{(x, y) \in \Omega_{r,c} \mid x^2 + y^2 - 1 = 0\}$ is $M = C$. If $x(t)^2 + y(t)^2 - 1 = 0 \ \forall t \geq 0$ then it follows by differentiating with respect to time that $x(t)\dot{x}(t) + y(t)\dot{y}(t) = 0$. Substituting $x(t)^2 + y(t)^2 - 1 = 0$ into the system's ODE gives:

$$\begin{aligned}\dot{x} &= \omega y \\ \dot{y} &= -\omega x,\end{aligned} \tag{1}$$

The above (linear) ODE also satisfies $x(t)\dot{x}(t) + y(t)\dot{y}(t) = 0$, thus it will trace out the same curve as the original ODE for the system. To see this, integrating both sides form 0 to t gives:

$$\begin{aligned}\int_0^t (x(\tau)\dot{x}(\tau) + y(\tau)\dot{y}(\tau))d\tau &= 0 \\ \int_0^t (x(\tau)dx(\tau) + y(\tau)dy(\tau)) &= 0 \\ x(t)^2 + y(t)^2 - x(0)^2 - y(0)^2 &= 0.\end{aligned}$$

So, we obtain $x(t)^2 + y(t)^2 = 1$ for all $t > 0$ if and only if $x(0)^2 + y(0)^2 = 1$. Therefore, the trajectories in I that satisfy $x(t)^2 + y(t)^2 = 1$ $\forall t \geq 0$ coincide with all those that satisfy $x(0)^2 + y(0)^2 = 1$. Therefore, $M = C$. By (i), (ii) and (iii) all trajectories starting in $\Omega_{r,c}$ converges to the set C.